

# Resume Builder — Technical Developer Guide

Complete architecture reference detailing data models, multi-step resume lifecycle, DOM canvas thumbnail pipelines, and the Google Gemini AI Generation & Smart Replay Engine.

React 18 & Vite

Node.js & Express

MongoDB & Mongoose

Google Gemini AI

Tailwind CSS

JWT & Bcrypt

html2canvas & html2pdf.js

## 1 🛠️ Tech Stack Overview

### ⚙️ Backend Architecture

- ▶ **Runtime & Framework:** Node.js, Express.js (ES Modules)
- ▶ **Database:** MongoDB with Mongoose ODM
- ▶ **AI Engine:** @google/genai SDK (Google Gemini)
- ▶ **Security & Auth:** JSON Web Token (JWT), BcryptJS password hashing
- ▶ **File Storage:** Multer disk storage (avatars & canvas snapshots)
- ▶ **Networking:** CORS enabled with static asset serving (/uploads)

### 🎨 Frontend & Client Application

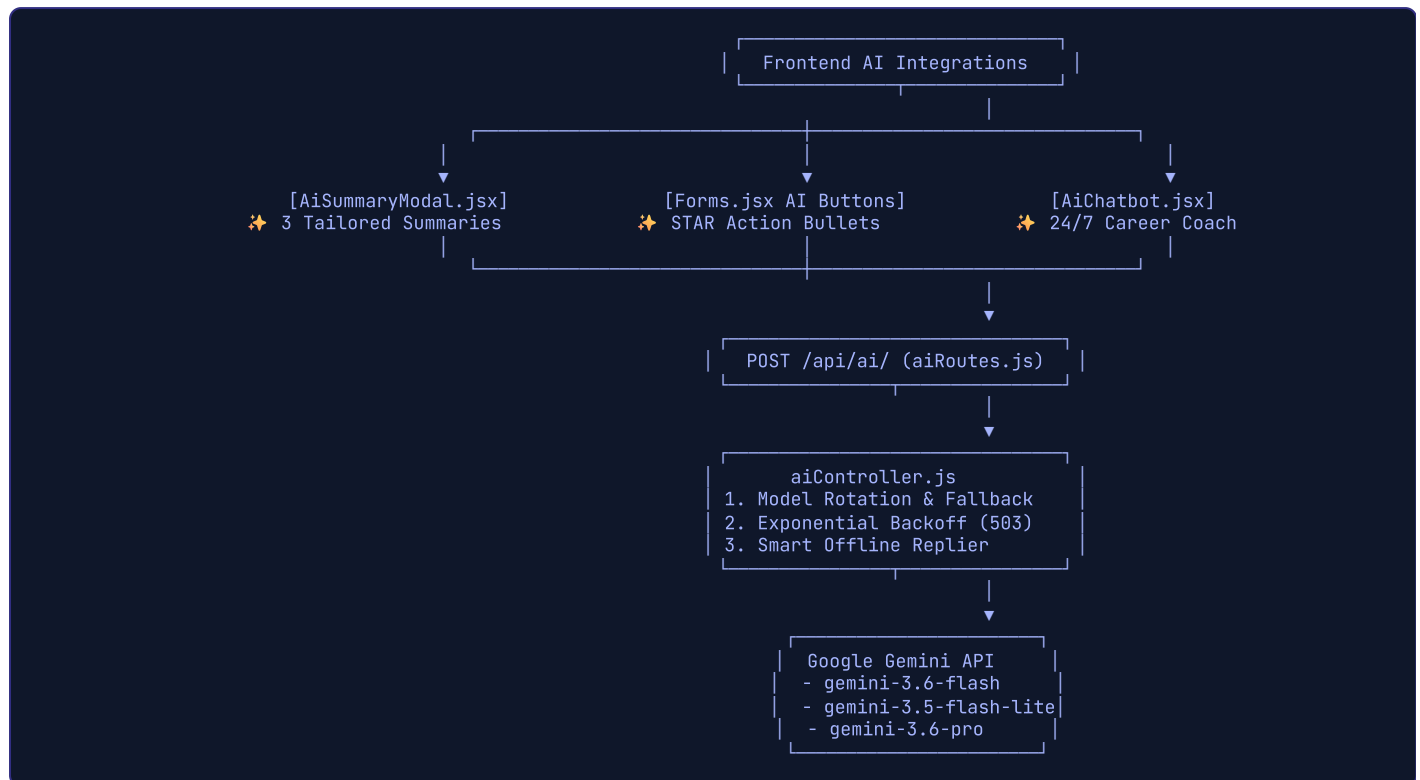
- ▶ **Framework & Bundler:** React 18, Vite (fast HMR)
- ▶ **Styling & Icons:** Tailwind CSS, Lucide React icons
- ▶ **Routing & State:** React Router DOM v6, React Context API
- ▶ **Network Client:** Axios with automatic Bearer token injection
- ▶ **Export & Capture:** html2canvas, html2pdf.js
- ▶ **Feedback:** react-hot-toast notifications

## 2 📁 Workspace & Folder Structure

```
Resume-2/
├── backend/
│   ├── config/
│   │   └── db.js # MongoDB connection logic
│   ├── controllers/
│   │   ├── aiController.js # AI Summary, STAR Bullets, & Chatbot Engine
│   │   ├── resumeController.js # CRUD operations for resumes
│   │   ├── uploadImages.js # Uploads profile pic & captured resume thumbnail
│   │   ├── userController.js # Register, Login, Get User Profile
│   │   ├── authMiddleware.js # JWT verification ('protect')
│   │   ├── uploadMiddleware.js # Multer diskStorage & file filters
│   │   ├── resumeModel.js # Comprehensive Resume schema
│   │   ├── userModel.js # User schema (name, email, password)
│   │   ├── aiRoutes.js # /api/ai endpoints
│   │   ├── resumeRoutes.js # /api/resume endpoints
│   │   ├── uploadRoutes.js # /api/upload endpoints
│   │   └── userRoutes.js # /api/auth endpoints
│   ├── middleware/
│   ├── models/
│   ├── routes/
│   ├── uploads/
│   ├── .env.example & server.js # Static directory storing uploaded images/thumbnails
│   └── # App entry point, CORS, static routes, DB init
├── frontend/src/
│   ├── context/
│   ├── pages/
│   │   ├── UserContext.jsx # Global auth state & token management
│   │   ├── Dashboard.jsx # User dashboard listing all saved resumes & creation modal
│   │   ├── LandingPage.jsx # Landing / marketing page
│   │   └── ResumeEditor.jsx # Editor shell & routing helper
│   ├── components/
│   │   ├── AiChatbot.jsx & AiSummaryModal.jsx # AI Assistant & Summary Generator
│   │   ├── EditResume.jsx & Forms.jsx # Master editor & sub-forms
│   │   ├── TemplateOne.jsx to TemplateSix.jsx # 6 Modern resume templates
│   │   ├── RenderResume.jsx, ThemeSelector.jsx, StepProgress.jsx, Cards.jsx, Navbar.jsx
│   └── utils/
│       ├── apiPaths.js, axiosInstance.js, color.js, helper.js
```

### 3 🤖 AI Generation & Replay Engine Architecture

Enterprise-grade AI engine built on top of the **Google Gemini SDK (@google/genai)** with candidate model rotation, exponential backoff, and intelligent offline response generators.



#### 1 AI Summary Generator

Generates 3 humanized, ATS-optimized summaries tailored by designation, level (fresher to executive), tone, and skills.

#### 2 STAR Bullet Enhancer

Transforms raw inputs into recruiter-ready bullets using Google's X-Y-Z formula with strong past-tense action verbs.

#### 3 24/7 AI Career Coach

Interactive floating chatbot with persona recognition, full-app guidance, skill suggestions, and interview preparation.

#### ★ Summary Response Structure

```
{
  "success": true,
  "data": {
    "summaries": [
      {
        "title": "Impact & Results-Driven",
        "summary": "Accomplished Full Stack Developer..."
      },
      {
        "title": "Technical & Skills-Focused",
        "summary": "Results-oriented Software Engineer..."
      },
      {
        "title": "Concise & Punchy",
        "summary": "Dedicated developer focused on clean code..."
      }
    ],
    "recommended": "Accomplished Full Stack Developer..."
  }
}
```

#### 🛡️ AI Fault-Tolerance & Safety Net

- ▶ **1. Candidate Model Rotation:** Tries **gemini-3.6-flash** first; if unavailable, automatically rolls over to **gemini-3.5-flash-lite** or **gemini-3.6-pro**.
- ▶ **2. Exponential Backoff:** Automatically pauses and retries on transient 503 (server overloaded) or 429 (rate limit) errors.
- ▶ **3. Smart Offline Replay Engine:** If API keys are missing or offline, the backend serves curated, high-quality level-specific summaries and bullets with 100% zero UI crashes.

 **User Model** ( backend/models/userModel.js )

| FIELD      | TYPE   | DETAILS                            |
|------------|--------|------------------------------------|
| name       | String | User display name (Required)       |
| email      | String | Unique user email (Required)       |
| password   | String | Bcrypt hashed password (Required)  |
| timestamps | Date   | Auto-managed createdAt & updatedAt |

 **Resume Model** ( backend/models/resumeModel.js )

Contains 12 comprehensive nested schemas for profile, education, work history, ratings, and design configurations.

userId

title

thumbnailLink

template

profileInfo

contactInfo

workExperience[]

education[]

skills[]

projects[]

certifications[]

languages[]

interests[]

| SECTION FIELD         | SUB-FIELDS & TYPE                                              | DESCRIPTION                                                               |
|-----------------------|----------------------------------------------------------------|---------------------------------------------------------------------------|
| userId                | ObjectId (ref: 'User')                                         | Owner reference for permissions and multi-user data isolation.            |
| title                 | String                                                         | Resume document title (e.g. "Senior Full Stack Engineer Resume").         |
| thumbnailLink         | String                                                         | Static URL path of the captured canvas preview snapshot.                  |
| template              | { theme: String, colorPalette: [String] }                      | Selected visual template design (1–6) and active hex color accents.       |
| profileInfo           | { fullName, designation, summary, profilePreviewUrl }          | Full name, target job role, ATS professional summary, and avatar link.    |
| contactInfo           | { email, phone, address, website, linkedin, github }           | Applicant contact points, physical city/state, and verified social links. |
| workExperience        | [{ companyName, jobTitle, startDate, endDate, description }]   | Job history with STAR-enhanced action bullet descriptions.                |
| education             | [{ institutionName, degree, startDate, endDate, description }] | Academic history, universities, degrees, and extracurriculars.            |
| skills                | [{ skillName: String, progress: Number }]                      | Technical proficiencies with 0–100 progress rating.                       |
| projects              | [{ title, description, githubLink, liveDemoLink }]             | Portfolio projects, repositories, live URLs, and impact statements.       |
| certifications        | [{ title: String, issuer: String, year: String }]              | Accreditations, professional licenses, and issuance years.                |
| languages & interests | [{ name, progress }], [String]                                 | Spoken language fluency ratings and professional interest tags.           |

## 5 Resume Lifecycle, Authentication & Environment

### 1 Step 1: Creation on Dashboard (POST /api/resume)

Initializes a MongoDB document with default structure & returns new `_id`.

### 2 Step 2: Multi-Step Editor Controller (EditResume.jsx)

8 sequential sub-forms with real-time completion score & AI helpers.

### 3 Step 3: Color Fix & Thumbnail Capture (uploadResumeImages)

Renders hidden DOM node, converts `oklch` colors, renders PNG via `html2canvas`.

### 4 Step 4 & 5: Upload, DB Update & PDF Export (html2pdf.js)

Multer saves snapshot, persists full JSON data, and exports pristine A4 vector PDF.

### 6. Authentication Flow

- ▶ **1. Auth:** `/api/auth/register` or `/login` issues signed JWT.
- ▶ **2. Client:** Saved in `localStorage.getItem('token')`.
- ▶ **3. Interceptor:** `axiosInstance.js` injects `Bearer <token>`.
- ▶ **4. Guard:** `authMiddleware.js` validates token and protects routes.

### 7. Environment Setup

```
PORT=5000
MONGO_URI=mongodb://127.0.0.1:27017/resume-builder
JWT_SECRET=your_jwt_secret_key
GEMINI_API_KEY=your_google_gemini_api_key
```

## 8 Complete API Reference Cheatsheet

| ENDPOINT                      | METHOD | AUTH   | PARAMETERS / PAYLOAD                       | DESCRIPTION                                        |
|-------------------------------|--------|--------|--------------------------------------------|----------------------------------------------------|
| /api/auth/register            | POST   | Public | { name, email, password }                  | Register new user account                          |
| /api/auth/login               | POST   | Public | { email, password }                        | Authenticate user & issue JWT                      |
| /api/auth/profile             | GET    | Bearer | -                                          | Get current logged-in user profile                 |
| /api/resume                   | POST   | Bearer | { title, ... }                             | Create a new resume document                       |
| /api/resume                   | GET    | Bearer | -                                          | Get all resumes belonging to user                  |
| /api/resume/:id               | GET    | Bearer | id in route param                          | Fetch specific resume by ID                        |
| /api/resume/:id               | PUT    | Bearer | Updated resume JSON                        | Update resume fields & completion score            |
| /api/resume/:id/upload-images | PUT    | Bearer | Multipart(thumbnail, profileImage)         | Upload thumbnail snapshot & avatar                 |
| /api/upload/image             | POST   | Bearer | Multipart(image)                           | Standalone image / photo upload                    |
| /api/resume/:id               | DELETE | Bearer | id in route param                          | Delete resume & cleanup static files               |
| /api/ai/generate-summary      | POST   | Bearer | { role, experienceLevel, tone, keySkills } | Generate 3 humanized ATS summaries via Gemini      |
| /api/ai/enhance-bullets       | POST   | Bearer | { role, text, type }                       | Convert experience/project into STAR bullets       |
| /api/ai/chat                  | POST   | Bearer | { message, history, resumeContext }        | Interactive AI Career & Resume Coach               |
| /uploads/<filename>           | GET    | Public | File static path                           | Serves uploaded images & thumbnails (CORS enabled) |